Research 1 — BASE RESEARCH

Crime Risk Prediction Incorporating Geographical Spatiotemporal Dependency into Machine Learning Models

Authors: Yue Deng, Rui-Xia He, Yang Liu

Published: October 2023, *Information Sciences*, Vol. 646, Article 119414

DOI: 10.1016/j.ins.2023.119414

Source: ScienceDirect / DOI

Research focus: The study incorporates **spatiotemporal lag variables** into machine-learning models and examines environmental and demographic factors affecting crime risk using Dallas crime data. The study reports that incorporating spatial-temporal lag variables can improve prediction accuracy and uses interpretable ML to examine variable contributions.

Gap identified: The research is based on Dallas and focuses on robbery/crime-risk prediction. It does not address **Tamil Nadu's heterogeneous district geography, population differences, multi-category crime structure and integrated evidence/provenance requirements**.

How ANVESHA fills the gap: ANVESHA adapts the **spatiotemporal dependency concept to Tamil Nadu districts**, incorporating neighboring-district patterns, historical trends, population-normalized crime rates, multiple crime categories and explainable predictions within one framework.

Why this is the BASE: This is the closest methodological foundation for ANVESHA because it connects **geography + time + ML + contextual variables + interpretability** in one journal study.

Research 2

Public Decision Support for Low Population Density Areas: An Imbalance-Aware Hyper-Ensemble for Spatio-Temporal Crime Prediction

Authors: Cristina Kadar, Rudolf Maculan, Stefan Feuerriegel

Published: April 2019, *Decision Support Systems*, Vol. 119, pp. 107–117

DOI: 10.1016/j.dss.2019.03.001

Source: ScienceDirect / DOI

Research focus: Spatiotemporal crime prediction in **low-population-density regions**, specifically addressing highly unequal crime distributions and class imbalance.

Gap identified: The study addresses sparsity and imbalance but does not provide a Tamil Nadu-specific framework combining **population normalization, district geography, multiple crime categories, explainability and data provenance**.

How ANVESHA fills the gap: ANVESHA explicitly analyses **low-count districts and rare crime categories**, uses population-adjusted crime rates, and integrates spatial/temporal information with provenance-aware data processing.

Research 3

CrimeTensor: Fine-Scale Crime Prediction via Tensor Learning with Spatiotemporal Consistency

Authors: Weichao Liang, Zhiang Wu, Zhe Li, Yong Ge

Published: March 2022, *ACM Transactions on Intelligent Systems and Technology*, Vol. 13, No. 2

DOI: 10.1145/3501807

Source: ACM / DOI

Research focus: Fine-scale crime prediction using the **spatial, temporal and categorical correlations** contained in crime incidents. The paper specifically addresses limitations of approaches that do not adequately model these three dimensions together.

Gap identified: The research concentrates on the prediction model and does not provide a broader **population-aware, explainable and provenance-driven crime intelligence framework**.

How ANVESHA fills the gap: ANVESHA adopts a **District × Time × Crime Category** representation and combines it with population normalization, spatial relationships, explainability, uncertainty and data provenance.

Research 4

Mapping the Risk Terrain for Crime Using Machine Learning

Authors: Andrew P. Wheeler, Wouter Steenbeek

Published: 2021, *Journal of Quantitative Criminology*, Vol. 37, pp. 445–480

DOI: 10.1007/s10940-020-09457-7

Source: Springer / DOI

Research focus: Uses machine learning and risk-terrain variables to predict crime at fine geographic scales. The study finds nonlinear relationships between environmental factors and crime and examines how ML predictions can be interpreted.

Gap identified: The study operates at a fine-scale US urban setting and does not address **state-wide Indian administrative districts, population normalization or multi-source provenance**.

How ANVESHA fills the gap: ANVESHA transfers the environmental/spatial-risk concept to **Tamil Nadu district-level analysis**, combining geographic, demographic and crime variables with population-adjusted rates and explainable predictions.

Research 5

Spatio-Temporal Crime Predictions by Leveraging Artificial Intelligence for Citizens Security in Smart Cities

Authors: Umair Muneer Butt, Sukumar Letchmunan, Fadratul Hafinaz Hassan, Mubashir Ali, Anees Baqir, Tieng Wei Koh, Hafiz Husnain Raza Sherazi

Published: March 2021, *IEEE Access*, Vol. 9, pp. 47516–47529

DOI: 10.1109/ACCESS.2021.3068306

Source: IEEE Access / DOI

Research focus: Uses **HDBSCAN** for hotspot detection and **SARIMA** for future crime prediction using ten years of New York City crime data.

Gap identified: The framework combines hotspot detection and time-series forecasting, but is primarily designed around an urban smart-city dataset and does not integrate **population-normalized district analysis, multi-source contextual data, explainability and provenance**.

How ANVESHA fills the gap: ANVESHA combines **hotspot/spatial analysis + temporal forecasting + population normalization + contextual data + explainable ML** for heterogeneous Tamil Nadu districts.

Research 6

Spatial-Temporal Meta-path Guided Explainable Crime Prediction

Authors: Yuting Sun, Tong Chen, Hongzhi Yin

Published: 2023, *World Wide Web*, Vol. 26, pp. 2237–2263

DOI: 10.1007/s11280-023-01137-3

Source: Springer / DOI

Research focus: Introduces a spatial-temporal meta-path framework that explicitly models relationships among **crime trends, environmental factors and social factors**, with a focus on explainable prediction.

Gap identified: The model focuses on complex urban relational data and does not address **Tamil Nadu's district-level administrative structure, population normalization or provenance-based evidence tracking**.

How ANVESHA fills the gap: ANVESHA incorporates **spatial relationships, environmental/social contextual variables and explainable predictions** within a Tamil Nadu district framework, while retaining the source and derivation history of the data.

Research 7

Information Fusion in Crime Event Analysis: A Decade Survey on Data, Features and Models

Authors: Kaixi Hu, Lin Li, Xiaohui Tao, Juan D. Velásquez, Patrick Delaney

Published: December 2023, *Information Fusion*, Vol. 100, Article 101904

DOI: 10.1016/j.inffus.2023.101904

Source: Information Fusion / DOI

Research focus: A decade review of **data fusion, feature fusion and model fusion** in crime-event analysis. It highlights the importance of combining environmental, spatial, temporal and other contextual information because crime cannot always be explained by historical crime data alone.

Gap identified: The paper is a systematic survey rather than a **Tamil Nadu-specific operational implementation** of a multi-source crime-intelligence pipeline.

How ANVESHA fills the gap: ANVESHA operationalizes multi-source fusion by integrating **crime records, population, geographic boundaries, area, density and contextual variables**, while recording data-source and transformation provenance.

Research 8

Machine Learning in Crime Prediction

Authors: Karabo Jenga, Cagatay Catal, Gorkem Kar

Published: 2 February 2023, *Journal of Ambient Intelligence and Humanized Computing*, Vol. 14, pp. 2887–2913

DOI: 10.1007/s12652-023-04530-y

Source: Springer / DOI

Research focus: Systematic review of **68 machine-learning crime-prediction studies**, examining datasets, features, algorithms, evaluation measures and research trends.

Gap identified: The literature demonstrates extensive use of ML for crime prediction but also shows that crime prediction involves diverse **spatial, temporal, socioeconomic and contextual variables**, with no single model universally solving the problem.

How ANVESHA fills the gap: ANVESHA uses a **model-comparison framework rather than relying on one algorithm**, and combines spatial, temporal, demographic and contextual features with explainability and provenance.

Research 9

A Systematic Review of Multi-Scale Spatio-Temporal Crime Prediction Methods

Authors: Yingjie Du, Ning Ding

Published: 23 May 2023, *ISPRS International Journal of Geo-Information*, Vol. 12, No. 6, Article 209

DOI: 10.3390/ijgi12060209

Source: MDPI / DOI

Research focus: Reviews crime prediction across different **spatial and temporal scales**, distinguishing micro-, meso- and macro-level prediction and short-, medium- and long-term forecasting.

Gap identified: Existing studies operate at different spatial and temporal scales, making direct comparison difficult. A major opportunity is to create a framework that consistently connects **district-level geography with multiple temporal scales**.

How ANVESHA fills the gap: ANVESHA establishes a consistent **Tamil Nadu district × time × crime-category framework**, allowing monthly/quarterly/annual patterns and spatial differences to be analysed within one system.

Research 10

Leveraging Transfer Learning with Deep Learning for Crime Prediction

Authors: Umair Muneer Butt, Sukumar Letchmunan, Fadratul Hafinaz Hassan, Tieng Wei Koh

Published: 17 April 2024, *PLOS ONE*, Vol. 19, Issue 4, e0296486

DOI: 10.1371/journal.pone.0296486

Source: PLOS ONE / DOI

Research focus: Addresses the problem of insufficient data for deep-learning crime prediction using **transfer learning and BiLSTM**, evaluated across Chicago, New York and Lahore crime datasets.

Gap identified: Transfer learning helps overcome limited data, but the framework does not integrate **Tamil Nadu district geography, population normalization, spatial dependencies, multi-source contextual evidence and provenance**.

How ANVESHA fills the gap: ANVESHA can evaluate transfer-learning or deep-learning approaches within Tamil Nadu while adding **district relationships, population-adjusted crime rates, contextual variables, explainability and evidence provenance**.
